

L-systems: Why Mathematical Modelling is Rooted in the Power of Abstraction

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June 11, 2026

1 A Model of Natural Growth

Mathematics began as a way to understand the world around us. A way to keep track and make sense of an ever-growing amount of information. The trade, farming, and architecture of the ancient world all required a method, or model, of quantifying reality in a meaningful way. Many of the nature elements of our reality are extremely complex, and so modelling them seems like an incredibly difficult task. For example, the striking visual uniqueness and variety of trees is enough to make us question if it is even possible at all. But by noticing the patterns, regularities and symmetries of the world, we can use mathematics to begin to build a model. By abstracting, something seemingly complex can become more understandable. Throughout the twentieth century, the L-system[1] became a simple mathematical tool to model the growth of plants, inspired by the self similar branching structure of algae (Figure 1).

But how far can we go? There is only so much you are able to include in a model; *George Box* famously argued that “All models are wrong, but some are useful”. When does abstracting venture so far as to remove vital detail. Can a simple system ever really capture the extreme complexity of trees, or even any chain reaction such as lightning (Figure 2)?

So how do we model growth? What are the systems limitations? Any why do models remain useful even when there are many omissions?



Figure 1: An image of fungal growth, an example of the filamentous organisms which inspired the creation of L-systems.



Figure 2: An example of sprite lightning, a natural phenomenon that has branch like properties.

1.1 The L-system

Throughout history humans have attempted to capture in mathematical elegance the beauty of nature[2]. The problem of trees begins with another question entirely: How can we model filamentous organisms (bacteria, fungi or algae)? The biologist Aristid Lindenmayer wrote his founding paper[1] on the exploration of this topic, laying the groundwork for growth models. He did this by creating a system that with each rendition, replaced all its parts based on a set of rules, taking advantage of the repetitive quality of branching.

Definition

Lindenmayer began by defining 3 things: First the alphabet V , a set of all symbols which were to be used in the production of the string which describes the growth behaviour. By then simply introducing a starting string or axiom ω , and a set of rules $p_1, p_2, \dots, p_k$ for how the elements of the string or substrings should change on each iteration n , a growing system was created. These now formalised string-based systems were later subsequently termed L-systems[2].

Example

Define an alphabet $V = \{A, B\}$, an axiom $\omega = A(n = 0)$, and two production rules:

$$p_1 : A \rightarrow AB, \quad p_2 : B \rightarrow A$$

Applying these rules in parallel at each iteration produces the following sequence of strings:

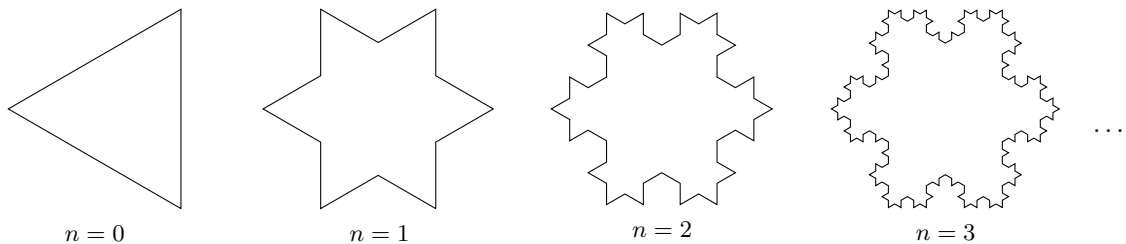
$n = 0$ A
 $n = 1$ AB
 $n = 2$ ABA
 $n = 3$ $ABAAB$
 $n = 4$ $ABAABABA$
 $n = 5$ $ABAABABAABAAB$

Notice that the lengths of these strings are 1, 1, 2, 3, 5, 8, 13...

... the Fibonacci sequence. A sequence of numbers which we see everywhere in nature and mathematics[3]. The appearance of the sequence, which is associated with the idea of geometric perfection in nature: the golden ratio[4], is a clear connection between mathematics and the natural idea of growth. This is the original example Lindenmayer used in his original 1968 paper[1] to model the growth of algae (as seen in figure 1), and it is already revealing: a two-rule system produces a sequence with deeply rooted mathematical and possibly natural structure.

1.2 The Origin of L-systems: The First Fractals and Formal Grammars

These new systems stemmed from a mathematical object called formal Grammars, conceived by Chomsky in the 1950s[5] when he applied the rewriting rules of the Kosh construction (seen below), a simple system with an initiator ($n = 0$) and a generator which produced the first fractals by replacing each line segment of the initiator with the generating chain repeatedly, to the features of natural languages. In this way, Chomsky set the initiator as a starting symbol instead of line, and the generator as a set of rules that replaced that symbol with a string of more symbols.



What changed?

The key difference between L-systems derived for modelling algae growth and these was that the productions in Chomsky Grammars are applied sequentially (meaning they are not deterministic because the order in which you apply the rules matters), whereas L-systems are done in parallel at every iteration. This distinction is also not just technical, it is biological: a developing cell does not divide one cell at a time, but many in unison. Lindenmayer built the mathematics to accurately represent this.

The result is a set of rules and starting conditions that when applied in parallel over many generations produce an exponential string growth with evolving complexity built in. Consequently, these strings became the cornerstone for the modelling of plants with Mathematics.

1.3 From Strings to Plants: The Turtle Interpretation

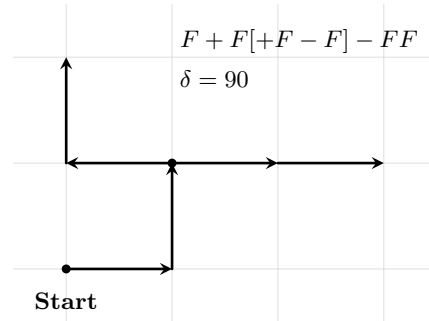
Now that one system is in place, we can model the idea of repetitive patterned growth with the L-system, but we have not yet captured the visual aspect of growth. To begin to model the plants

that we see around us, a more sophisticated and consistent graphical application of L-systems is needed. To do this, the computer scientists Szilard and Quinton adapted the known 'turtle' method of generating computer graphics to visualize the abstract rewriting rules[6].

The idea of the turtle interpretation is to define the state of a turtle as a triplet (x, y, α) where (x, y) are the coordinates or position and α is the angle or heading of the turtle, the way it is facing. With a given step size d and angle increment δ we can now create an alphabet where each symbol corresponds to a specific action on the turtle's state[7]. When used alongside the set of rules of an L-system, the resulting string represents the graph of the movement of the 'turtle' in space. For our purposes, we can define the following alphabet:

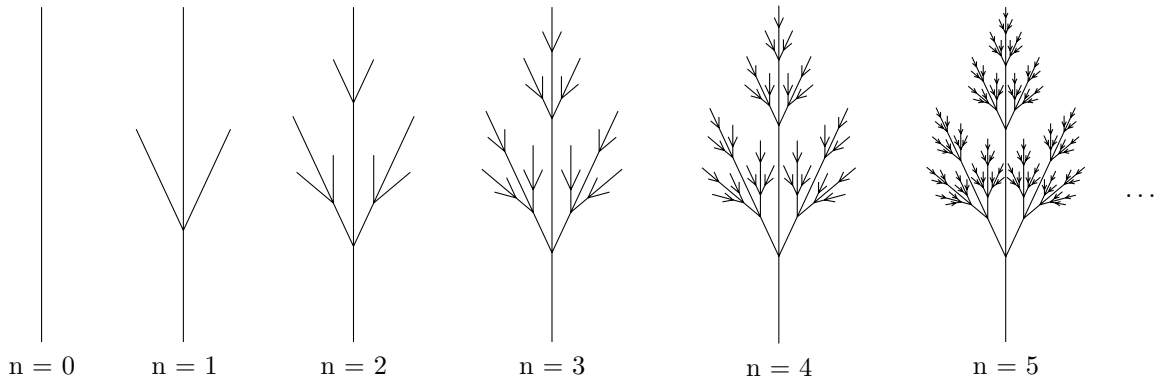
- F** move forward by d and draw a line: $(x, y, \alpha) \rightarrow (x + d \cos \alpha, y + d \sin \alpha, \alpha)$
- +** turn left by δ : $(x, y, \alpha) \rightarrow (x, y, \alpha + \delta)$
- turn right by δ : $(x, y, \alpha) \rightarrow (x, y, \alpha - \delta)$
- [** Save the current state of the turtle to the stack
-]** Move the turtle back to the last saved state and remove it from the stack

Using these rules on a grid, you can imagine the baby turtle shuffling forward on the beach, turning left and right, scrambling to reach the ocean, being picked up and dropped back down by vicious seagulls, where the path drawn out can be thought of as the string of instructions for the turtle.



Now, with both an understanding of the graphical implementation and the recursive strings made by L-systems, growth can be visualised as elegant geometry. As seen below, the self-similar growth of the string encoded in geometric fractals takes a simple set of starting conditions and creates something both visually and Mathematically beautiful:

$$\omega: X \quad p_1: X \rightarrow F[+X][-X]FX \quad p_2: F \rightarrow FF \quad \delta = 25^\circ$$

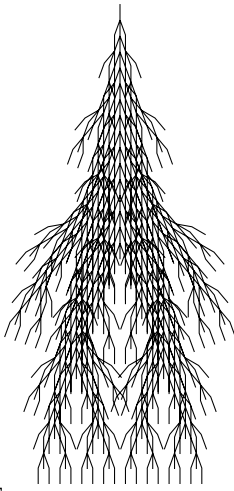


With the corresponding strings:

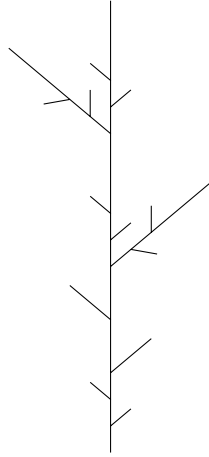
- $n = 0$ X
- $n = 1$ $F[+X][-X]FX$
- $n = 2$ $FF[+F[+X][-X]FX][-F[+X][-X]FX]FFF[+X][-X]FX$
- $n = 3$ $FFFF[+FF[+F[+X][-X]FX][-F[+X][-X]FX]FFF[+X] \dots 143 \text{ characters total}$
- $n = 4$ 451 characters
- $n = 5$ 1,391 characters

Now that both systems are set up and working together and we can in one sense model anything that ‘grows’. That is, the L-system gives us a way to imitate the repeating generation of new branching structures using a set of simple rules, while the turtle interpretation grounds these abstract strings in geometry. In unison, they give us a visual representation of the structure of trees, organisms, networks and chain reactions.

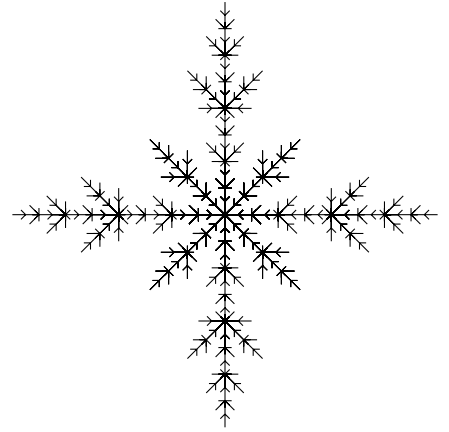
The main strength of the model is its simplicity. Originally designed by Lindenmayer to imitate the expansion of filamentous organisms such as the algae *Anabaena catenula*[1], the minimalist nature of the procedure allows it to be applied to a wide variety of natural phenomena with ease. Not only can it be used to model the growth of plants but also other chain reactions such as lightning and river deltas or biological systems such as neurons in a brain or vascular networks. Below are implementations of L-systems that capture the visual essence of sprite lightning (as seen in figure 1), bamboo, a snowflake, a crop, a bush and a branch.



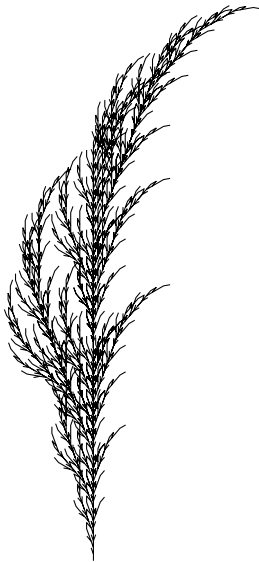
$\omega: F$
 $p_1: F \rightarrow F[+F - F][-F + F]F$
 $n = 4, \delta = 20$



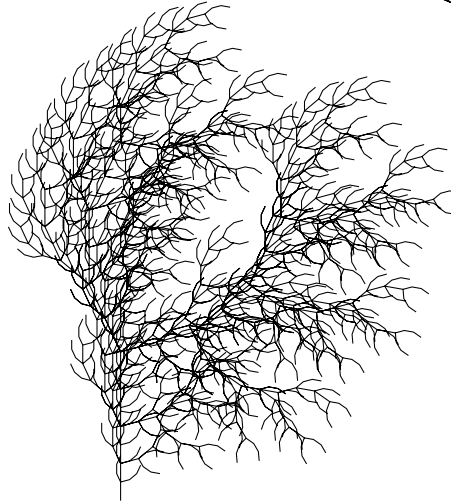
$\omega: F$
 $p_1: F \rightarrow F[-X]X[+X]X$
 $p_2: X \rightarrow FX$
 $n = 4, \delta = 50$



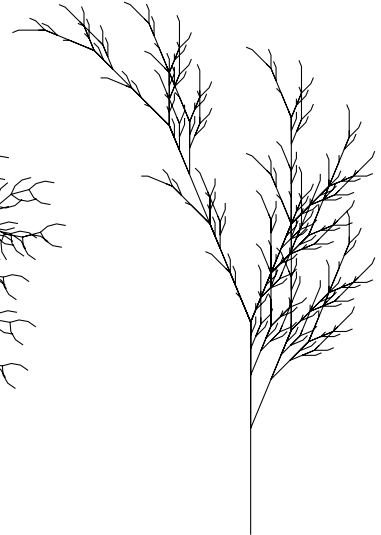
$\omega:$
 $[F][- - F][+ + F][+ + + F]$
 $p_1: F \rightarrow F[-F][+F]FF$
 $n = 5, \delta = 45$



$\omega: F$
 $p_1: F \rightarrow F[+F]F[F - F]$
 $n = 5, \delta = 20$



$\omega: F$
 $p_1: F \rightarrow FF - [- - F + F + F] + [+F - F - F]$
 $n = 4, \delta = 25$



$\omega: X$
 $p_1: X \rightarrow F - [[X] + X] + F[+FX] - X$
 $p_2: F \rightarrow FF$
 $n = 5, \delta = 22.5$

1.4 Where the Model Falls Short

However, like all Mathematical models of the world, a solution so elegant purposefully leaves things out. A comprehensive system of would become so convoluted and complex and more and more factors were introduced to the point where it would no longer be productive to use.

Physics

Perhaps the most obvious oversight is that the model does not exist in the real world like trees do. L-systems do not consider any Physics, unlike in real life, these topological objects do not respond to gravity, wind, or mechanical stress. In a similar way, the process of growth is not continuous as it is with reality, but discrete and unchanging, simply reiterating the same rules over and over instead of adapting to its surroundings. This leads to the next problem; real plants respond and interact with an environment. Sunlight, soil, competition and relationships with other organisms all also come into play in a organisms development[2].

Natural Laws

Further, these systems do not obey principles of nature such as the optimisation principle[8] that showed that real branching networks obey scaling laws derived from minimising energy dissipation. The $\frac{3}{4}$ power law for metabolic rate emerges from fractal vascular geometry. Standard L-systems have no such constraint built in; they can produce biologically impossible branching ratios.

Reversibility

A complete system would also be reversible, meaning given a real tree there would be a systematic way to derive the L-system that generated. For these reasons, L-systems are less a model of natural growth, and more a model of self similarity in branching structures that can be interperated as plant growth to interigate curtain aspects of development.

2 Why does this “wrong” model remain one of the most powerful tools in mathematical biology?

However, despite how easy it is to scrutinise, these limitations are not failures of mathematics, but choices about what we want to model. Returning to Box’s timeless quote “All models are wrong, but some are usefull” [9], the model is in many senses an inaccurate image of reality, it lacks physics, does not obey natural laws and is irreversible, but it is useful for just as many reasons. Its success is in the real insights it gives us in such simplicity.

We could add physics, but then it would take too long to compute graphics. We could introduce the limitations of biological laws, but then would would lose the option to explore different chain reactions that are not biological such as lightning or river deltas. In this way, the genius of a simple system that can generate complex results is that it provides and extremely solid framework to build specialised systems from.

Power Law

For example, West[8] used the fractal dimension that emerges from branching network theory, the same theory that underpins L-systems, to derive the $\frac{3}{4}$ power law. The law states that an organims metabolic rate scales to the $\frac{3}{4}$ power of its body mass. This is true from bacteria to blue whales. It is surprising that the metabolic rate is not proportional to mass (M^1) because the implications are huge; an elephant is 1000 times more massive than a mouse, but only uses 178x more energy. The law was hypothesised prior by imperical evidence, but without the abstraction of metabolic systems as branching fractals, it may never have been mathematically derieved. The same geometric insights and self similar heirachy are the basis of the L-system, and in this way, the model may not be not exactly accurate to reality, but evidently can still be used extremely effectively to understand the underlying structure of the world through simplification.

*Botanical
Classifica-
tion*

L-systems can also serve as a tool for understanding botanical classification[2]. L-systems can effectively capture within their ruleset the characteristics of different species, flowering patterns, and branching structures; allowing us to compare similar specimens topologically. These distinctions can describe how different plants allocate reasources and prioritize elements of growth, giving us mathematical classifications for natural strategies of survival and growth such as:

Monopodial: A single main stem with lateral branches extending out from it in various ways, optimised to avoid self-shading and maximise light capture, often resulting in a conical shape like a pine tree (figure 3).

Sympodial: Multiple stems that grow from a common termination of the main axis (often in a flower) and continue outwards (figure 4).



Figure 3: The Monopodial growth of a Norfolk Island Pine.



Figure 4: The Sympodial growth strategy of an English Oak tree.

These strategies are not only visual, but distinctly indicative of how different plants allocate resources. A monopodial plant invests more energy into the main stem to reach high and avoid self-shading or reach above competitors in a dense environment such as the evergreen forests of the northern hemisphere[10]. On the other hand, a sympodial plant splits out from the main stem to maximise surface area and light capture from all directions, typical of a plant in a more open environment. These insights, derived from the topological structure of the plant, can be used to understand and justify the evolutionary strategies of different plants, and the environments they are found in. So, while the L-system may not be a perfect model of the process of the growth of a tree, it may be used to understand the underlying reasons for the specifics of that growth.

*Computer
Graphics*

In another field, the topological essence of growth described by L-systems and the Turtle interpretation is also useful for computer graphics[7]. This is because it does not matter if the visuals are physically accurate, they just have to look correct. The simplicity of the L-system allows us to create realistic models of plants and trees in an accurate and variable way. This has applications in video games, animation rendering, and art; where apparent complexity is desired but would otherwise be tedious to create by hand. Further, the introduction of probabilistic rules and variation into the L-system (stochastic L-systems) allows for the condensed implementation of variety and randomness, as is true of real plant growth. For the same reason, the simplicity of the L-systems allows a large amount of variation to be stored without the need for a large amount of data, which is ideal for computer graphics where a large amount of diversity is needed in a small amount of space. Overall, the fact that the L-systems is not accurate is not a drawback for graphics, but a strength; it allows for more flexibility, control, and variety in less space.

2.1 The usefulness of being wrong

In conclusion, L-systems are sequences of symbols that are generated by applying a set of rules recursive process that give rise to complex structures, resembling what we understand as growth. They are powerful because they take advantage of the self-similarity of the natural world to encode complexity into a small amount of data. L-systems fall short at representing reality entirely because they fail to take into account the multitude of factors that influence real world growth.

However, this is a strength in disguise. The surprising visual and dimensional accuracy achieved with such a simple set of rules allows for implementations across disciplines, from video game graphics to botanical classification. The way which the model captures the underlying topological structure may seem at first abstract, but is genuinely revealing of universal properties of biological networks in the real world.

So, the main insight L-systems is that nature is not devoid of simplicity but built on it. The fact that what is at first glance undecipherable and intricate, can be modeled in its essence by breaking it down into simpler parts, is the everpresent tension between effectiveness and accuracy in Mathematical modelling. Return to Box's quote, "All models are wrong, but some are useful"[9], the L-system is a perfect example of this. It does not capture or take into consideration physics, the environment, or even adhere to natural laws, but it still maintains usefulness. The fact that Aristid Lindenmayer's initial implementation of rewriting rules to model the growth of algae can

do anything from capture the visual essence of a tree to inspire the derivation of a fundamental observed law in biology, is a testament to the power of being ‘wrong’.

While the Mathematical models of our world are not perfect at capturing reality, they are fantastic tools for understanding it. This is the fundamental usefulness of abstraction in Mathematical modelling, revealed by the L-system.

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